
Relational Abstraction

Lecture 7 — Static Analysis by Abstract Interpretation

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From intervals to relational domains

Closure operators and abstract domains

Closure of difference-bound matrices

The domain of convex polyhedra

§ 1

From intervals to relational domains

Intervals forget relations

Consider

```
x := 0; y := 10;  
while x < 10 do  
  x := x + 1; y := y - 1
```

At the loop head, narrowing can recover only

$$x \in [0, 10], \quad y \in [-\infty, 10].$$

But intervals have forgotten the invariant $x + y = 10$.

At loop exit, $x = 10$, while the interval for y remains $[-\infty, 10]$. The analyzer cannot prove that $y = 0$.

Missing information

Bounds on individual variables do not record dependencies between variables.

Zones: difference constraints

A *zone* is a conjunction of constraints

$$x - y \leq c, \quad x, y \in \text{Var} \cup \{0\}, \quad c \in \mathbb{Z} \cup \{+\infty\},$$

where the special variable 0 always has value zero.

This single form expresses:

$$x \leq u \quad \text{as} \quad x - 0 \leq u,$$

$$x \geq \ell \quad \text{as} \quad 0 - x \leq -\ell,$$

$$x = y \quad \text{as} \quad x - y \leq 0 \wedge y - x \leq 0.$$

Zones retain every interval bound and additionally relate pairs of variables.

Difference-bound matrices

A zone is represented by a *difference-bound matrix* (DBM): entry M_{xy} stores the constraint $x - y \leq M_{xy}$.
The zone $0 \leq x = y \leq 10$ is represented by

M_{xy}	0	x	y
0	0	0	0
x	10	0	0
y	10	0	0

For example, $M_{x0} = 10$ means $x \leq 10$, while $M_{xy} = M_{yx} = 0$ means $x = y$.

Concretization

$$\gamma_Z(M) \triangleq \{\sigma \mid \sigma(0) = 0 \wedge \forall u, v, \sigma(u) - \sigma(v) \leq M_{uv}\}.$$

An entry $+\infty$ means that no constraint is recorded for this pair.

Exercise — build a difference-bound matrix

Give the DBM over the variables $0, x, y$ representing the conjunction

$$x \in [-1, 3] \quad \wedge \quad x - y \leq 2 \quad \wedge \quad y - x \leq 4.$$

Use $+\infty$ when no constraint is given.

Exercise — build a difference-bound matrix

Solution

$$M = \begin{array}{c|ccc} & 0 & x & y \\ \hline 0 & 0 & 1 & +\infty \\ x & 3 & 0 & 2 \\ y & +\infty & 4 & 0 \end{array}$$

Each entry M_{uv} is read directly from the bound $u - v \leq M_{uv}$.

Tightening a zone with one constraint

For one difference constraint $q \equiv u - v \leq c$, define

$$\text{tighten}(M, q)_{rs} \triangleq \begin{cases} \min(M_{uv}, c) & (r, s) = (u, v), \\ M_{rs} & \text{otherwise.} \end{cases}$$

Thus only the entry representing $u - v$ may become tighter.

Filtering by a single difference constraint will be the basic case of $\text{filter}_Z^\#$.

Soundness

For $q \equiv u - v \leq c$,

$$\gamma_Z(\text{tighten}(M, q)) = \llbracket u - v \leq c \rrbracket \gamma_Z(M) = \{\sigma \in \gamma_Z(M) \mid \sigma(u) - \sigma(v) \leq c\}.$$

Indeed, every entry is unchanged except M_{uv} , which becomes $\min(M_{uv}, c)$: satisfying this new bound is equivalent to satisfying both the old bound and q .

In the equality case, write $M_1 = \text{tighten}(M, u - v \leq c)$. For $\bowtie \in \{\leq, <, =, \geq, >\}$, define

$$\text{atom}_Z^\#(p, M) \triangleq \begin{cases} \text{tighten}(M, u - v \leq c) & p = (u - v \leq c), \\ \text{tighten}(M, u - v \leq c - 1) & p = (u - v < c), \\ \text{tighten}(M_1, v - u \leq -c) & p = (u - v = c), \\ \text{atom}_Z^\#(v - u \leq -c, M) & p = (u - v \geq c), \\ \text{atom}_Z^\#(v - u < -c, M) & p = (u - v > c), \\ \text{atom}_Z^\#(u - 0 \bowtie c, M) & p = (u \bowtie c), \\ M & p \text{ is unsupported.} \end{cases}$$

For an unsupported atom, returning M loses precision but is sound: no possible state is removed.

The complete zone filter

After pushing negations down to comparisons, define

$$\text{filter}_Z^\#(b, M) \triangleq \begin{cases} M & b = \text{true}, \\ \perp_Z & b = \text{false}, \\ \text{atom}_Z^\#(p, M) & b = p, \\ \text{filter}_Z^\#(b_2, \text{filter}_Z^\#(b_1, M)) & b = b_1 \wedge b_2, \\ \text{filter}_Z^\#(b_1, M) \sqcup_Z \text{filter}_Z^\#(b_2, M) & b = b_1 \vee b_2. \end{cases}$$

Exercise — filtering a difference-bound matrix

Let M be the DBM computed in the exercise on Slide 5:

$$M = \begin{array}{c|ccc} & 0 & x & y \\ \hline 0 & 0 & 1 & +\infty \\ x & 3 & 0 & 2 \\ y & +\infty & 4 & 0 \end{array}$$

Compute $\text{filter}_Z^\#(x = y, M)$ and identify the entries that change.

Exercise — filtering a difference-bound matrix

Solution

$$\text{filter}_Z^\#(x = y, M) = \text{tighten}(\text{tighten}(M, x - y \leq 0), y - x \leq 0)$$

$$= \begin{array}{c|ccc} & 0 & x & y \\ \hline 0 & 0 & 1 & +\infty \\ x & 3 & 0 & 0 \\ y & +\infty & 0 & 0 \end{array}$$

Only M_{xy} and M_{yx} change: both are tightened to 0.

Assigning in a zone: $x := y + k$

For $x := y + k$, substitute $y + k$ for the new value of x . The resulting matrix M' is defined by

$$M'_{uv} \triangleq \begin{cases} 0 & u = v = x, \\ M_{yv} + k & u = x, v \neq x, \\ M_{uy} - k & u \neq x, v = x, \\ M_{uv} & u \neq x, v \neq x. \end{cases}$$

Example 1

Suppose the input zone entails $x = y$, represented by $M_{xy} = M_{yx} = 0$. After the assignment $x := x + 1$,

$$M'_{xy} = 1, \quad M'_{yx} = -1,$$

which expresses $x = y + 1$.

The complete zone assignment

Write $\text{affine}(x, y, k, M)$ for the preceding transformer implementing $x := y + k$. Then

$$\text{assign}_Z^\#(x, a, M) \triangleq \begin{cases} \text{affine}(x, 0, k, M) & a = k, \\ \text{affine}(x, y, 0, M) & a = y, \\ \text{affine}(x, y, k, M) & a = y + k \text{ or } a = k + y, \\ \text{affine}(x, y, -k, M) & a = y - k, \\ \text{forget}(x, M) & \text{otherwise.} \end{cases}$$

The fallback is defined entrywise by

$$\text{forget}(x, M)_{uv} \triangleq \begin{cases} 0 & u = v = x, \\ +\infty & (u = x) \text{ xor } (v = x), \\ M_{uv} & u \neq x \wedge v \neq x. \end{cases}$$

It discards all information about the new value of x , but never retains obsolete constraints from before the assignment.

Exercise — assigning in a zone

Consider the DBM representing $0 \leq x = y \leq 10$:

$$M = \begin{array}{c|ccc} & 0 & x & y \\ \hline 0 & 0 & 0 & 0 \\ x & 10 & 0 & 0 \\ y & 10 & 0 & 0 \end{array}$$

Compute the DBM obtained after the assignment $x := y + 1$.

Exercise — assigning in a zone

Solution

$$\text{assign}_Z^\#(x, y + 1, M) = \begin{array}{c|ccc} & 0 & x & y \\ \hline 0 & 0 & -1 & 0 \\ x & 11 & 0 & 1 \\ y & 10 & -1 & 0 \end{array}$$

The result represents $0 \leq y \leq 10$, $1 \leq x \leq 11$, and $x = y + 1$.

A first zone analyzer

- atomic guards update one matrix entry;
- assignments $x := y + k$ update the row and column of x ;
- unsupported assignments may safely forget the row and column of their target;
- join takes the pointwise maximum of two matrices;
- widening is still needed: the zone domain also has infinite ascending chains.

§2

Closure operators and abstract domains

Why canonical abstract descriptions matter

A concretization need not be injective:

$$a \neq b \quad \text{but} \quad \gamma(a) = \gamma(b).$$

Consequences

- syntactic equality does not mean semantic equality;
- order tests may miss concrete inclusion;
- transfer functions accumulate redundant information;
- fixed-point detection may be delayed.

Closure and reduction operators provide exactly this normalization.

Desired invariant

Choose one canonical representative for each concretization:

$$\gamma(a) = \gamma(b) \implies a = b.$$

We want an equivalent domain in which γ is injective.

Upper closure operators

Let $(C, \sqsubseteq_C)$ be a complete lattice.

Definition 2

An *upper closure operator* is a function $\rho : C \rightarrow C$ satisfying

$$c \sqsubseteq_C \rho(c), \quad c \sqsubseteq_C d \implies \rho(c) \sqsubseteq_C \rho(d), \quad \rho(\rho(c)) = \rho(c).$$

Its fixed points are the properties representable without further loss:

$$\text{Fix}(\rho) \triangleq \{c \in C \mid \rho(c) = c\}.$$

Example 3

The interval hull is an upper closure on $\mathcal{P}(\mathbb{Z})$:

$$\rho_I(S) \triangleq \gamma_I(\alpha_I(S)).$$

Applying the interval hull twice changes nothing.

Closed elements form a complete lattice

Let $A_\rho \triangleq \text{Fix}(\rho)$, ordered as in C .

Theorem 4

A_ρ is a complete lattice.

For every family $(a_i)_{i \in I}$ of closed elements:

$$\bigsqcap_{i \in I}^{A_\rho} a_i = \bigsqcap_{i \in I}^C a_i, \quad \bigsqcup_{i \in I}^{A_\rho} a_i = \rho \left(\bigsqcup_{i \in I}^C a_i \right).$$

Why is the concrete meet already closed?

If $m = \bigsqcap_i a_i$, monotonicity gives $\rho(m) \sqsubseteq_C \rho(a_i) = a_i$ for every i , hence $\rho(m) \sqsubseteq_C m$. Extensivity gives the reverse inequality.

Thus a closure operator selects a meet-closed family of concrete properties, also called a *Moore family*.

A Galois connection induces a closure

Let

$$C \overset{\alpha}{\underset{\gamma}{\rightleftarrows}} A$$

be a Galois connection, and define

$$\rho \triangleq \gamma \circ \alpha : C \rightarrow C.$$

1. Sound abstraction gives $c \sqsubseteq_C \gamma(\alpha(c)) = \rho(c)$.
2. Monotonicity of α and γ makes ρ monotone.
3. Since $\alpha \circ \gamma \sqsubseteq_A \text{id}_A$,

$$\rho(\rho(c)) = (\gamma \circ \alpha \circ \gamma \circ \alpha)(c) \sqsubseteq_C (\gamma \circ \alpha)(c) = \rho(c).$$

Extensivity applied to $\rho(c)$ gives the reverse inequality.

Hence ρ is an upper closure operator. Moreover, every $\gamma(a)$ is closed; conversely, $\rho(c) = c$ implies $c = \gamma(\alpha(c))$. Therefore

$$\text{Fix}(\rho) = \gamma(A).$$

The closed concrete properties are exactly those described by the abstract domain.

A closure induces a Galois insertion

Conversely, start from an upper closure $\rho : C \rightarrow C$ and let

$$A_\rho \triangleq \text{Fix}(\rho), \quad \alpha_\rho(c) \triangleq \rho(c), \quad \gamma_\rho(a) \triangleq a.$$

For $c \in C$ and $a \in A_\rho$:

$$\alpha_\rho(c) \sqsubseteq_C a \iff \rho(c) \sqsubseteq_C a \iff c \sqsubseteq_C a \iff c \sqsubseteq_C \gamma_\rho(a).$$

Therefore

$$C \begin{matrix} \xrightarrow{\alpha_\rho} \\ \xleftarrow{\gamma_\rho} \end{matrix} A_\rho$$

is a Galois connection. Since γ_ρ is inclusion, $\alpha_\rho \circ \gamma_\rho = \text{id}_{A_\rho}$: it is a Galois insertion.

Equivalence

Abstract domains up to isomorphism are equivalently described by upper closure operators on the concrete lattice.

Reducing a non-injective abstract domain

Given a Galois connection, define on the abstract side

$$\kappa \triangleq \alpha \circ \gamma : A \rightarrow A.$$

It is a *lower closure*, or *kernel operator*:

$$\kappa(a) \sqsubseteq_A a, \quad \kappa \text{ is monotone}, \quad \kappa(\kappa(a)) = \kappa(a).$$

Moreover,

$$\gamma(\kappa(a)) = \gamma(a).$$

Restricting to $A_{\text{red}} \triangleq \text{Fix}(\kappa)$ removes redundant representations. If $a, b \in A_{\text{red}}$, then

$$\gamma(a) = \gamma(b) \implies a = \alpha(\gamma(a)) = \alpha(\gamma(b)) = b.$$

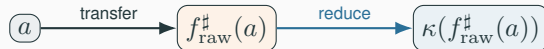
Thus concretization becomes injective on the reduced domain.

The role of reduction in an analyzer

Let $f_{\text{raw}}^\# : A \rightarrow A$ be a sound abstract transformer. Normalize its output:

$$f_{\text{red}}^\# \triangleq \kappa \circ f_{\text{raw}}^\#.$$

Because $\gamma \circ \kappa = \gamma$, reduction does not change the represented concrete property.



Practical benefits

Canonical values make equality, inclusion, join, and fixed-point stabilization reflect semantic properties rather than accidental syntax.

§3

Closure of difference-bound matrices

Raw DBMs are not canonical

Recall that M_{uv} represents $u - v \leq M_{uv}$. Consider the two matrices:

$$M = \begin{array}{c|ccc} & 0 & x & y \\ \hline 0 & 0 & 0 & 0 \\ x & +\infty & 0 & 3 \\ y & 2 & +\infty & 0 \end{array} \quad N = \begin{array}{c|ccc} & 0 & x & y \\ \hline 0 & 0 & 0 & 0 \\ x & 5 & 0 & 3 \\ y & 2 & 2 & 0 \end{array}$$

In M , the recorded constraints already imply

$$x \leq y + 3 \leq 5, \quad y - x \leq 2.$$

Therefore

$$M \neq N \quad \text{but} \quad \boxed{\gamma_Z(M) = \gamma_Z(N)}.$$

Syntactic matrix equality cannot yet test semantic equality of zones.

A DBM is a weighted graph

The entry $M_{uv} = c$ is an edge

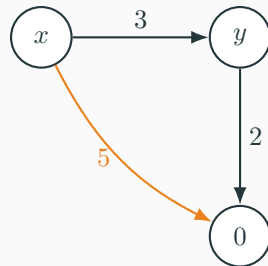
$$u \xrightarrow{c} v$$

expressing $u - v \leq c$.

Along a path,

$$u - v \leq a, \quad v - w \leq b \quad \implies \quad u - w \leq a + b.$$

The strongest implied bound is therefore the weight of a shortest path.



$$x - y \leq 3, \quad y \leq 2 \quad \implies \quad x \leq 5.$$

Exercise — draw the graph of a DBM

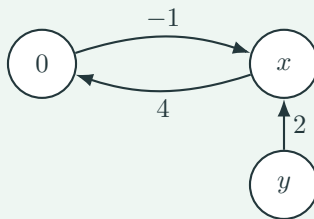
Give the weighted graph represented by

$$M = \begin{array}{c|ccc} & 0 & x & y \\ \hline 0 & 0 & -1 & +\infty \\ x & 4 & 0 & +\infty \\ y & +\infty & 2 & 0 \end{array}$$

Omit diagonal self-loops and entries equal to $+\infty$.

Exercise — draw the graph of a DBM

Solution



The finite off-diagonal entries give the edges $0 \xrightarrow{-1} x$, $x \xrightarrow{4} 0$, and $y \xrightarrow{2} x$.

Define the closure by all-pairs shortest paths:

$$\text{close}(M)_{uv} \triangleq \min\{\text{weight}(\pi) \mid \pi \text{ is a path from } u \text{ to } v\}.$$

Floyd–Warshall computes it by repeatedly applying

$$M_{uw} \leftarrow \min(M_{uw}, M_{uv} + M_{vw})$$

for every intermediate variable v .

Inconsistent zones

A negative cycle derives $u - u < 0$. Equivalently, $\text{close}(M)_{uu} < 0$ for some u ; the represented zone is empty.

Closure removes no concrete state: it only makes all consequences of the current constraints explicit.

DBM closure is a reduction

For every consistent DBM M :

$$\text{close}(M) \sqsubseteq_{\text{DBM}} M, \quad \gamma_Z(\text{close}(M)) = \gamma_Z(M), \quad \text{close}(\text{close}(M)) = \text{close}(M).$$

Closure is also monotone, so it is a lower closure operator on DBM representations.

Theorem 5 (Canonical DBMs)

For consistent DBMs,

$$\gamma_Z(M) = \gamma_Z(N) \iff \text{close}(M) = \text{close}(N).$$

Thus closed DBMs provide one representative per nonempty zone. They repair the lack of injectivity of the raw concretization.

Exercise — close a DBM

Starting from

$$M = \begin{array}{c|ccc} & 0 & x & y \\ \hline 0 & 0 & 0 & 0 \\ x & +\infty & 0 & 3 \\ y & 2 & +\infty & 0 \end{array},$$

compute the two finite bounds that replace $+\infty$ in $\text{close}(M)$. Explain each bound by a path.

Starting from

$$M = \begin{array}{c|ccc} & 0 & x & y \\ \hline 0 & 0 & 0 & 0 \\ x & +\infty & 0 & 3 \\ y & 2 & +\infty & 0 \end{array},$$

compute the two finite bounds that replace $+\infty$ in $\text{close}(M)$. Explain each bound by a path.

Solution

$$\text{close}(M)_{x0} = M_{xy} + M_{y0} = 3 + 2 = 5, \quad \text{close}(M)_{yx} = M_{y0} + M_{0x} = 2 + 0 = 2.$$

Hence $\text{close}(M)$ is the matrix N from the preceding example.

Close after abstract operations

Let the raw transformers be those from Lecture 6. Define

$$\begin{aligned}\text{filter}_Z^\#(b, M) &\triangleq \text{close}(\text{filter}_{Z,\text{raw}}^\#(b, M)), \\ \text{assign}_Z^\#(x, a, M) &\triangleq \text{close}(\text{assign}_{Z,\text{raw}}^\#(x, a, M)), \\ M \sqcup_Z N &\triangleq \text{close}((\max(M_{uv}, N_{uv}))_{uv}).\end{aligned}$$

On closed DBMs, semantic inclusion becomes an exact matrix test:

$$\boxed{\gamma_Z(M) \subseteq \gamma_Z(N) \iff \forall u, v, M_{uv} \leq N_{uv}.}$$

Equality and loop stabilization can now be checked structurally without missing equivalent representations.

Exercise — closure preserves useful information

Start from the DBM M of the preceding exercise, which represents

$$x \geq 0, \quad y \geq 0, \quad x - y \leq 3, \quad y \leq 2,$$

and analyze the sequence

$$y := 0; \quad x := x + 1.$$

Task

Compute the final intervals for x and y

1. starting directly from the raw DBM M ;
2. starting from the closed DBM $\text{close}(M)$.

Which implied bound is preserved by closure?

Exercise — closure preserves useful information

Start from the DBM M of the preceding exercise, which represents

$$x \geq 0, \quad y \geq 0, \quad x - y \leq 3, \quad y \leq 2,$$

and analyze the sequence

$$y := 0; \quad x := x + 1.$$

Solution

Without closure, assigning $y := 0$ discards the path $x \xrightarrow{3} y \xrightarrow{2} 0$. The result is

$$x \in [1, +\infty], \quad y = 0.$$

Closure first records the implied bound $x \leq 5$. It survives the assignment to y , so after $x := x + 1$ the result is

$x \in [1, 6], \quad y = 0.$

§ 4

The domain of convex polyhedra

Convex polyhedra

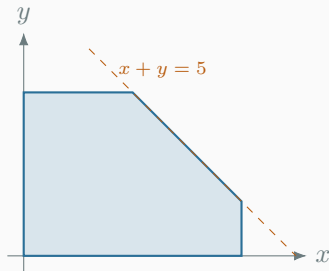
A *convex polyhedron* in $\mathbb{Q}^n$ is the solution set of finitely many linear inequalities:

$$P = \{x \in \mathbb{Q}^n \mid Ax \leq b\}.$$

Polyhedra express arbitrary linear relations, for example

$$0 \leq x \leq 4, \quad 0 \leq y \leq 3, \quad x + y \leq 5.$$

Unlike zones, coefficients are not restricted to differences of two variables.



A lattice of polyhedra

Let Poly_n be the set of convex polyhedra in $\mathbb{Q}^n$, ordered by inclusion. Include $\emptyset$ and $\mathbb{Q}^n$.

For two polyhedra:

$$P \sqcap Q = P \cap Q, \quad P \sqcup Q = \text{conv}(P \cup Q).$$

Both results are again polyhedra. Hence

$$(\text{Poly}_n, \subseteq, \cap, \text{conv}(\cup), \emptyset, \mathbb{Q}^n)$$

is a lattice.

Polyhedra do not form a complete lattice

Consider in $\mathbb{Q}^2$ the rational points of the closed unit disk

$$D = \{(x, y) \in \mathbb{Q}^2 \mid x^2 + y^2 \leq 1\}.$$

For $n \geq 2$, let P_n be the rational points of the regular 2^n -gon centered at the origin and circumscribed about D . These polygons can be chosen so that

$$P_2 \supseteq P_3 \supseteq P_4 \supseteq \cdots, \quad D = \bigcap_{n \geq 2} P_n.$$

Every P_n is a polyhedron, but their intersection D is not a polyhedron: no finite set of linear inequalities describes a circle.

Therefore Poly_2 is not closed under arbitrary meets and is not a complete lattice.

When does a concretization admit an abstraction?

Theorem 6

Let C and A be complete lattices and $\gamma : A \rightarrow C$ monotone. An abstraction $\alpha : C \rightarrow A$ making (α, γ) a Galois connection exists exactly when γ preserves arbitrary meets, including the empty meet:

$$\gamma\left(\bigsqcap_A S\right) = \bigsqcap_C \{\gamma(a) \mid a \in S\}.$$

When this condition holds, the only possible abstraction is

$$\alpha(c) \triangleq \bigsqcap \{a \in A \mid c \sqsubseteq_C \gamma(a)\}.$$

Indeed, c is a lower bound of all $\gamma(a)$ occurring in the set above. Meet preservation therefore gives $c \sqsubseteq_C \gamma(\alpha(c))$. Then

$$\alpha(c) \sqsubseteq_A a \iff c \sqsubseteq_C \gamma(a).$$

Why there is no polyhedral abstraction function

Let D be the rational points of the closed unit disk:

$$D = \{(x, y) \in \mathbb{Q}^2 \mid x^2 + y^2 \leq 1\}.$$

Then $\alpha_P(D)$ would have to be the least polyhedron containing D . But no such polyhedron exists: a geometric argument shows that every polygon containing D can be transformed into a strictly smaller polygon that still contains D .

Thus the disk has sound polyhedral over-approximations, but no best one. A global abstraction function α_P cannot be defined.

The absence of a global best abstraction does not prevent sound analysis. For each concrete operation, derive a polyhedral result whenever its image happens to be polyhedral.

Local exactness

If $f(\gamma_P(P))$ is itself a polyhedron, define

$$f^\sharp(P) \triangleq f(\gamma_P(P)).$$

The transformer is then exact for this operation, without requiring α_P on arbitrary concrete sets.

Otherwise, choose some sound polyhedral over-approximation. There may be no most precise choice.

The cases that work exactly

Let $P \in \text{Poly}_n$.

Operation	Exact polyhedral transformer
linear guard $a \cdot x \leq c$	$P \cap \{x \mid a \cdot x \leq c\}$
affine assignment $\mathbf{x} \leftarrow T\mathbf{x} + d$	$\{T\mathbf{x} + d \mid \mathbf{x} \in P\}$
forget variables	linear projection of P
join	$\text{conv}(P \cup Q)$

Half-space intersection, affine image, projection, and binary convex hull all preserve polyhedra. Nonlinear guards and assignments generally leave the domain and require approximation.

Exercise — an exact polyhedral transfer

Initially,

$$P = \{(x, y) \mid 0 \leq x \leq 2, y = x\}.$$

Compute the polyhedron after

$$x := x + 1; \quad \text{filter}(x + y \leq 3).$$

Exercise — an exact polyhedral transfer

Initially,

$$P = \{(x, y) \mid 0 \leq x \leq 2, y = x\}.$$

Compute the polyhedron after

$$x := x + 1; \quad \text{filter}(x + y \leq 3).$$

Solution

After the assignment:

$$x - y = 1, \quad 0 \leq y \leq 2.$$

The guard gives $(y + 1) + y \leq 3$, hence $y \leq 1$. Therefore

$$P' = \{(x, y) \mid x - y = 1, 0 \leq y \leq 1\}.$$

Two representations of a polyhedron

H-representation

Intersection of half-spaces:

$$P = \{x \mid Ax \leq b\}.$$

Convenient for:

- guards and intersection;
- inclusion tests;
- detecting inconsistency.

V-representation

Vertices, rays, and lines:

$$P = \text{conv}(V) + \text{cone}(R) + \text{span}(L).$$

Convenient for:

- affine images;
- projection;
- convex hull.

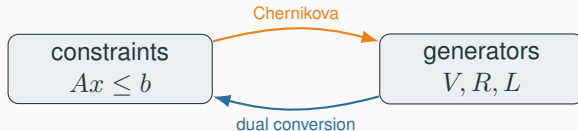
A practical polyhedral library needs to move between both descriptions.

Chernikova's double-description method

Chernikova's algorithm converts between the two representations of a polyhedron.

For $V = \{v_i\}$, $R = \{r_j\}$, and $L = \{\ell_k\}$,

$$P = \text{conv}(V) + \text{cone}(R) + \text{span}(L)$$
$$= \left\{ \sum_i \lambda_i v_i + \sum_j \mu_j r_j + \sum_k \nu_k \ell_k \mid \begin{array}{l} \lambda_i, \mu_j \in \mathbb{Q}_{\geq 0}, \quad \nu_k \in \mathbb{Q}, \\ \sum_i \lambda_i = 1 \end{array} \right\}.$$



- For intersection, use the H-representations, then convert the result to a V-representation.
- For convex hull, use the V-representations, then convert the result to an H-representation.

Precision has a computational cost

Polyhedra can express arbitrary conjunctions of linear inequalities, but:

- the number of constraints and generators may grow exponentially;
- conversion between H- and V-representations can be exponential;
- redundant constraints and generators must be controlled;
- widening remains necessary for loops.

Domain-design trade-off

intervals $\longrightarrow$ zones $\longrightarrow$ polyhedra

gains relational precision while increasing computational cost.

Takeaways

1. Closure and reduction provide canonical representations without changing their meaning.
2. Zones express difference constraints; polyhedra express arbitrary linear inequalities.
3. Polyhedra provide more precision at a greater computational cost, but admit no global best abstraction.